Ravishankar Anirudh | Doctoral student

ravishankar@mpia.de 0000-0002-3071-3365 rdotanirudh.github.io SciX Library

Research highlights

October 2023 – present
- **Ultra-high-resolution cosmological zoom-in simulations of galaxies at $z > 5.5$**
  Core member of the VESTRAL project. Contributed to the analysis of simulation outcomes across its development. *Preparing a manuscript*, with a focus on the surprisingly early and rapid formation of discs in low-mass ($M_* < 10^9\,\mathrm{M}_\odot$) galaxies in the first billion years.

October 2024 – June 2026
- **Mock JWST observations of $z = 2 - 6$ galaxies using TNG50**
  Created the first realistic mock JWST NIRCam and NIRSpec/MSA observations of Hα-emitting gas, for a sample of >4000 star-forming galaxies with stellar masses $> 10^8\,\mathrm{M}_\odot$. *Resulted in a publication in MNRAS.*

May 2022 – June 2025
- **Synthetic CO observations of $z = 1 - 3$ galaxies using Simba**
  Predicted the resolved CO emission of a sample of >1000 star-forming galaxies with stellar masses $> 10^9\,\mathrm{M}_\odot$. Co-developed the radiation prescriptions of the SLICK forward modelling pipeline. *Published a first-author paper and a co-author paper in A&A and ApJ.*

Science Interests

- Cosmological hydrodynamical simulations of high-redshift galaxies
- Disc formation, gas kinematics and dynamics
- Bursty star formation in galaxy discs at $z > 5.5$
- Comparing simulations to observations via forward modelling of emission from gas and stars, photometry and spectroscopy

Education

October 2023 – 2027 (expected)
- **Doctoral student of Astronomy and Cosmic Physics**
  at the Max Planck Institute for Astronomy / IMPRS-HD /Universität Heidelberg, Germany
  Supervisor: Dr. Annalisa Pillepich
  Thesis title (tentative): *Kinematics and galaxy growth in the early Universe: insights from cosmological simulations*

August 2018 – July 2023
- **Bachelor of Science - Master of Science Dual Degree** in Physics (GPA: 8.6)
  at the Indian Institute of Science Education and Research (IISER), Tirupati, India
  Supervisors: Dr. Melanie Kaasinen and Dr. Gergö Popping (ESO, Garching)
  Thesis title: *Using radiative transfer models and cosmological simulations to map the emission of cold gas within galaxies* (GPA: 9.3).

Mentoring Experience

Co-supervision

July – August 2026
- **Summer internship** of Michelle Park (MPIA)
  Project title: *Interactions drive bursty, spatially clustered star formation in VESTRAL*

February – July 2026
- **Master thesis project** of Jehanne Delhomelle (MPIA)
  Thesis title: *The effects of galaxy interactions on star formation at cosmic noon with TNG50*

Tutoring

July – August 2026
- **Laboratory course tutor** at Universität Heidelberg
  Supervised the wavefront analysis experiment for 6 groups of undergraduate students

September 2024 and 2025
- **Postgraduate course tutor** at Universität Heidelberg *(twice)*
  Supervised the Cosmology block course tutorial sessions for Master students

Selected Conferences and Workshops

Contributed Talks

September 2026 • **BUilding Galaxies from Scratch (BUGS) II**, Heidelberg
Title: *The emergence of disks during the Epoch of Reionization in cosmological dwarf galaxy simulations*

June – July 2026 • **European Astronomical Society Annual Meeting**, Lausanne *(three talks)*
Title 1: *Resolving stochastic star formation in cosmological dwarf galaxy simulations*
Title 2: *Predicting the resolved CO emission of $z = 1 - 3$ star-forming galaxies*
Title 3: *Testing the inference of kinematics from mock JWST NIRSpec/MSA observations*

June 2026 • **Zoom-in views of galaxy disks - Nordita workshop**, Stockholm *(two talks)*
Selected title: *A tale of early disks: Micro-labels and micro-galaxies*

June 2024 • **The feedback-driven matter cycle in galaxies**, Hengstberger Symposium, Heidelberg
Title: *Simulating bursty star formation in distant galaxies*

Collaboration Meeting Talks

May 2025 • **RUBIES Meeting**, Bergen

July 2025 • **CONDOR Meeting**, Heidelberg

Contributed Posters

September 2026 • **BUilding Galaxies from Scratch (BUGS) II**, Heidelberg

August 2024 • **Observing and Simulating Galaxy Evolution in the Era of JWST**, Ascona

July 2024 • **European Astronomical Society Annual Meeting**, Padova *(two posters)*

February 2024 • **BUilding Galaxies from Scratch (BUGS) I**, Vienna

Workshops and summer schools

November 2024 • **Heidelberg Graduate School For Physics Winter School**, Obergurgl

May 2024 • **IMPRS Hackathon**, Heidelberg

July-August 2021 • **European Southern Observatory Summer Research Programme**, Garching (remote)

Observing Proposals

- **JWST Cycle 6**: Co-I on a proposal (submitted) targeting ∼350 galaxies at z ∼4-7 using NIRSpec/MSA slit-stepping

Technical Skills

Codes • AREPO, Python, bash, git, HPC Slurm, LaTeX, C++ (basic), Fortran (basic), Html, css (basic)

Simulations • IllustrisTNG, VESTRAL, Simba

Astro. software • ADQL, astropy, CAESAR, Despotic, DS9, FITS, h5py, msafit, photutils, pysersic, TOPCAT, yt

Instruments • JWST NIRCam and NIRSpec/MSA forward modelling, Gaia, JCMT SCUBA-2, ZTF

Academic and Community Service

September 2026 • **Local Organizing Committee Member** of the "BUGS II" conference in Heidelberg

August 2024 – Present • **Working Conditions workgroup Coordinator** of the PhDNet of the Max Planck Gesellschaft, Germany

October 2024 – September 2026 • **External Student Representative** of the Max Planck Institute for Astronomy

October 2024 – September 2025 • **Internal Student Representative** of the Max Planck Institute for Astronomy

References

Dr. Annalisa Pillepich
Research Group Leader
Max Planck Institute for Astronomy
pillepich@mpia.de

Dr. Anna de Graaff
Lise Meitner Group Leader
Max Planck Institute for Astronomy
degraaff@mpia.de

Dr. Melanie Kaasinen
DECRA Fellow and Senior Lecturer
Australian National University
melanie.kaasinen@anu.edu.au